

Reviewer 1

Comments

Reviewer 1 - Initial submission, 14/09/2026

This paper presents TCAS, a training-free schedule for multi-view diffusion that balances structural alignment and texture quality. The authors report improved PSNR and texture retention relative to aggressive uniform scaling, alongside favorable overall human preference.

The revision strengthens the submission through cluster-aware preference analysis, investigation of adapter-dependent failure modes, and transparent reporting of unsuccessful FAC experiments. The method is promising, but the following concerns should be addressed before publication:

- Figures 4 and 6 show pronounced purple color shifts, repeated patterns, and substantial discrepancies from the reference appearance. Please provide full-object comparisons across a broader set of objects, including the unmodified pipeline and a competitive fixed-scale baseline. Baked meshes rendered from unseen viewpoints and an assessment of seams or cross-view consistency would help establish practical value. The supplementary video could not be assessed.
- Tables 4–6 evaluate the main adapter on 300 objects, including the 24 schedule-selection objects; Tables 7–8 use subsets of that pool. Table 9 uses the probe-disjoint 276-object subset but changes the adapter and competing schedules. It therefore does not directly confirm the main adapter’s reported 0.96 dB gain outside the selection set. Please evaluate the main adapter and relevant baselines under the same held-out protocol.
- The paper acknowledges PSNR’s limitations and includes perceptual evaluations. However, the connection between the metric gains and faithful texture reproduction remains unclear given the visible artifacts. Please distinguish texture variation from texture fidelity: larger gradients or Laplacian variance can reflect unwanted artifacts as well as meaningful detail.
- A confidence interval crossing zero for the FG-SSIM difference does not establish equivalence or non-inferiority. Table 4 also shows lower Edge-SSIM for TCAS than for aggressive scaling, 0.529 versus 0.559. Please report paired intervals, acknowledge this trade-off, and either justify a non-inferiority margin or moderate the preservation claim.
- Section 4.1.3 provides useful details, but it is still hard to reproduce the FAC experiments. The code is encouraged to be released,

TCAS is a potentially useful, low-cost intervention. Consistent held-out comparisons, clearer metric interpretation, and stronger qualitative evidence are needed to establish its practical benefit for 3D texturing.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 24/09/2026

Summary

This paper studies inference-time adapter scaling in geometry-conditioned multi-view diffusion for 3D texture generation. The key observation is that uniformly increasing the adapter scale improves structural metrics such as FG-SSIM, but may suppress texture synthesis, leading to flattened appearance, reduced color variation, or other high-frequency degradation. Based on this observation, the paper proposes Timestep-Conditioned Adapter Scaling (TCAS), which adopts a low-high-low schedule over the early, middle, and late denoising stages, with the selected configuration $C3=(1.25, 2.50, 1.25)$. The authors further introduce a Correction-Alignment-Interference (CAI) stage-utility analysis to motivate this schedule, and evaluate the method on a 24-object probe set, a disjoint 276-object holdout set, and a pooled 300-object evaluation set. The revised manuscript is substantially more careful than the earlier version in terms of experimental protocol, statistical analysis, and claim scope. In particular, the authors now explicitly separate the 24-object probe set from a strictly disjoint 276-object holdout for schedule transfer, provide more detailed implementation information, use cluster-aware statistics for the user study, and re-examine the learned FAC extension under a stricter paired and disjoint protocol. However, I still have significant concerns regarding the methodological novelty, the interpretation of the CAI formulation, and the lack of cross-backbone validation. The revision makes the empirical claims considerably more trustworthy, but it does not fundamentally change the nature of the contribution: TCAS remains a manually discretized three-stage inference schedule identified within one MVPainter-style architecture family.

Strengths

First, I appreciate that the revised version substantially improves the experimental rigor. The authors explicitly disclose that the 24-object probe set is part of the 300-object pool and re-evaluate competing schedules on the strictly disjoint 276-object holdout. This is a meaningful improvement over evaluating schedule-selection conclusions only on the same pool used for tuning. The holdout transfer experiments against linear warm-up, cosine bump, trapezoid, and Gaussian-peak schedules are useful and make the empirical comparison more convincing.

Second, the paper identifies a practically relevant failure mode. The observation that stronger geometry-conditioned residual injection may improve structural alignment while simultaneously degrading texture richness is useful for practitioners working with ControlNet-/adapter-like multi-view texture pipelines. The use of multiple texture diagnostics rather than a single structural metric is also appropriate.

Third, I appreciate the authors' treatment of the FAC extension. In the previous version, FAC was presented as a learned extension that improved over TCAS. In the revised manuscript, the authors re-evaluate FAC under a stricter paired and disjoint setting and report that none of the trained FAC variants outperforms the training-free schedule. The paper now clearly states that TCAS remains the main and sole method.

Major Concerns

1. The core methodological contribution remains quite limited. I am not yet convinced that this constitutes a sufficiently substantial methodological contribution beyond schedule selection for an existing adapter.
2. The CAI formulation appears to formalize an empirically selected schedule rather than derive it independently.
3. Cross-backbone generalization remains the most important missing experiment. The main experiments are conducted within a single MVPainter-style architecture family. Section 4.7 expands the analysis to adapters with different residual magnitudes and scale

semantics, and shows three possible high-scale behaviors: texture flattening, high-frequency artifact amplification, or negligible effect. This analysis is interesting, but it does not substitute for evaluation on a genuinely different geometry-conditioned multi-view diffusion backbone.

Reviewer 3

Comments

Reviewer 3 - Initial submission, 28/09/2026

In this revision, the authors have paid efforts in addressing the concerns in the last round. The revision now looks good for publication.